

STATISTICAL INFERENCE

Statistical Inference involves using data from a sample to make generalizations or predictions about a larger population.

Features of Inferential Statistics

1. **Population vs. Sample:**
 - **Population:** The entire group you want to study (e.g., all customers of a store).
 - **Sample:** A smaller, representative group taken from the population (e.g., 100 customers surveyed).
2. **Goals:**
 - **Estimation:** Estimate population parameters (e.g., mean, proportion) using sample statistics.
 - **Hypothesis Testing:** Test assumptions or claims about a population using sample data. (e.g., "Is the average customer satisfaction score above 4?").
3. **Tools and Methods:**
 - **Confidence Intervals:** A range of values likely to contain the population parameter (e.g., "The average height is between 160 cm and 170 cm with 95% confidence").
 - **Hypothesis Testing:** Testing a claim about the population using sample data (e.g., t-tests, chi-square tests).
 - **ANOVA (Analysis of Variance):** Comparing means across multiple groups.

Parameter:

A measurable characteristics of a population is called parameter. It is used to describe the entire population.

Eg. Population mean(μ), population variance(σ^2), population standard deviation (σ)

Statistic:

A measurable characteristics of a sample is called a statistics.

Eg: Sample mean ($\bar{x}$), sample variance (s^2), sample standard deviation(s)

In Inferential statistics we use sample statistics to estimate the corresponding population parameter.

TYPES OF INFERENTIAL STATISTICS

There are mainly 3 types of statistical inferences.

1. Point Estimation
2. Interval Estimation
3. Hypothesis Testing

HYPOTHESIS TESTING

A hypothesis is an assumption or claim about a population parameter.

Hypothesis testing is a statistical method used to make decisions or inferences about a population based on sample data. It involves formulating two competing hypothesis: the null hypothesis H_0 , and alternate hypothesis H_a .

Example: H_0 : Mean blood pressure of treated patients = untreated patients; H_1 : Mean blood pressure differs between groups.

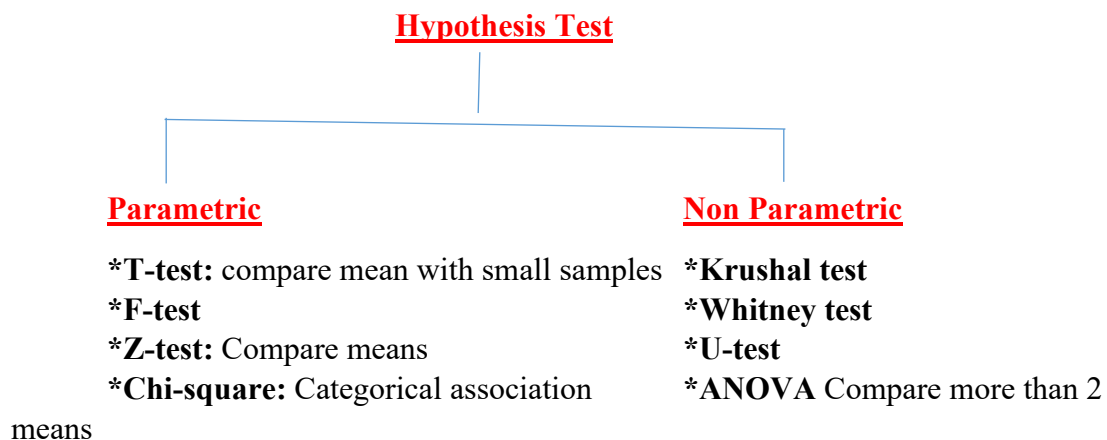
- The goal of Hypothesis Testing is to assess the evidence against null hypothesis and decide whether to reject it.
- The alternative hypothesis can take different forms, depending on the research question and the specific claim being tested.

We assign a **confidence level** to hypothesis testing and try to limit the amount of error being committed. The Universally accepted confidence level is 95%. By doing so we admit that while rejecting null hypothesis, there is 5% possibility of wrongly rejecting the null hypothesis.

TYPES OF HYPOTHESIS TESTING

Hypothesis testing methods are broadly divided into:

1. **Parametric Tests** : These tests assumes that the data follows a specific distribution, usually **normal distribution**. It uses actual numerical values i.e. parameters like: **Mean (μ)**, **Standard deviation (σ)**
2. **Non-Parametric Tests**: These tests do not assume any specific distribution. They often use Ranks, Signs, Frequencies instead of actual values and works well with skewed or ordinal data. They make use of data distribution to comment on the claim.



The main difference is based on the **assumptions made about the data**.

Difference between Parametric and Non-Parametric Tests

Feature	Parametric Test	Non-Parametric Test
Distribution assumption	Assumes normal distribution	No distribution assumption
Uses	Mean and SD	Ranks or signs
Data type	Interval/Ratio	Ordinal/Nominal
Sensitivity	More sensitive	Less sensitive
Power	Higher if assumptions hold	Lower compared to parametric
Outlier effect	Strongly affected	Less affected
Sample size	Usually larger	Works well for small samples

Common situations for Hypothesis testing:

$$H_0: \mu \leq \mu_0$$

$$H_a: \mu > \mu_0$$

$$H_0: \mu \geq \mu_0$$

$$H_a: \mu < \mu_0$$

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

The equality sign should appear on the null hypothesis side.

1. Testing Research Hypothesis:

Testing a Research Hypothesis involves systematically evaluating a proposed idea or claim about a population, based on sample data with evidence.

Example:

Suppose a motorbike has an average fuel efficiency of 40 km/L. R&D department of that company developed a new fuel injection system to increase the average fuel efficiency. Here research hypothesis is that new fuel injection system will have an average fuel efficiency exceeding 40km/L.

So Null hypothesis $H_0: \mu \geq 40$, alternative hypothesis $H_a: \mu < 40$

Here the alternative hypothesis is based on the assumption that research hypothesis is true.

2. Testing the Validity of a Claim:

Suppose a manufacturer of soft drinks claims that two –litre soft drink bottles has a low sugar level of not more than 100g sugar. So testing is done to test the manufacturer's claim.

Null Hypothesis $H_0: \mu \leq 150$

Alternative Hypothesis $H_a: \mu > 150$

The null hypothesis is based on the assumption that claim is true.

3. Testing in Decision Making Situations

This situation occurs when a decision maker must choose between two courses of action, one associated with null hypothesis and other associated with the alternative hypothesis. For example, on the basis of a sample of parts from a shipment just received, a quality control inspector must decide whether to accept the shipment or to return the shipment to the supplier as it does not meet the specifications, the mean length of 3 inch per part.

Null Hypothesis $H_0 : \mu = 3$

Alternative hypothesis : $\mu \neq 3$

Important terms in Hypothesis Testing

1. Null Hypothesis (H_0):

- The default assumption or claim we aim to test.
- Example: "The average recovery time for patients is 8 days."

2. Alternative Hypothesis (H_a):

- The claim we are trying to support if the null hypothesis is rejected.
- Example: "The average recovery time for patients is not 8 days."

3. Significance Level (α):

- It is the probability of rejecting the null hypothesis when it is actually true (making Type I error). It is commonly set at 0.05 (5%).
- Significance level = 1 – confidence level.

If significance level is 0.01, confidence level is 99%

If significance level is 0.05, confidence level is 95%

If significance level is 0.10, confidence level is 90%

4. Test Statistic:

- A value calculated from the sample data using z-test, t-test, chi square test to determine whether to reject the null hypothesis.
- Choose the test based on data type, distribution and sample size.

5. p-Value: (probability-value)

- The probability of getting the observed results (or something more extreme) if the null hypothesis is true.
- Lower the p-value strong evidence against H_0 .

Suppose a friend says: "I can throw a basketball into the hoop almost every time."

You watch them throw 20 times and they miss 18 shots.

Now you think: "If they were really that good, how likely is it to see such poor performance?"

- If this is **very unlikely**, you stop believing the claim.
- If this is **fairly possible**, you may still believe them.

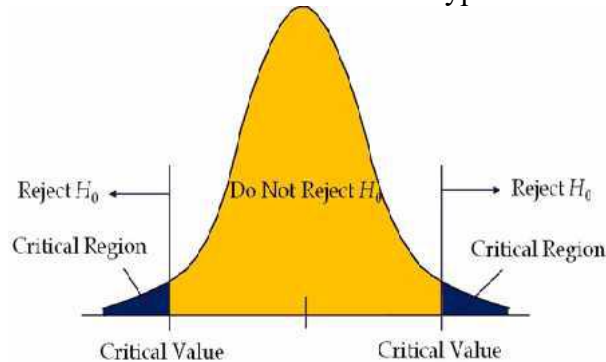
That “how unlikely” part is the p-value.

If $p\text{-value} \leq \alpha$ reject H_0 , or if $p\text{-value} > \alpha$ fail to reject H_0 .

6. Decision:

- Based on the p-value or critical value, decide whether to reject or fail to reject H_0

7. **Critical Region:** Critical regions, also known as rejection regions, are specific regions of test statistics distribution where the null hypothesis is rejected.



8. **Types of Errors:** In hypothesis testing we decide about a **null hypothesis** H_0 using sample data. Because we decide based on a sample, we can be wrong in two main ways.

Type I and Type II errors

Hypothesis is:	H_0 true	H_a true
Accept H_0	*Correct conclusion *True negative *Probability = $1 - \alpha$	*Type-II error *False negative Probability = β
Reject H_0	*Type-I error *False positive *probability = α	*Correct onclusion *True positive *probability = $1 - \beta$

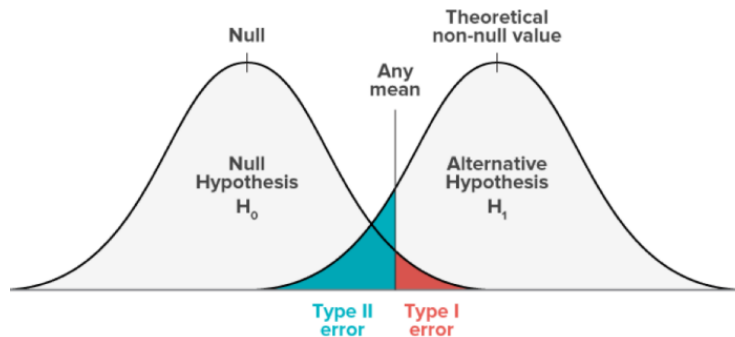
Type I error (α error / false positive)

- Definition: Rejecting H_0 when H_0 is actually true. Its probability is denoted by α (level of significance), often 0.05.
- Example: A Test says the person has disease even though the person is actually healthy.
- A test with a 95% confidence level means that there is a 5% chance of getting a type I error.

Type II error (β error / false negative)

- Definition: Failing to reject H_0 when H_0 is actually false. Its probability is β .
- Example: Concluding there is “no difference” between two teaching methods when in reality one method truly is better.

- **Power** of a test is $1 - \beta$ = probability of correctly rejecting a false H_0
- We make Type- II error by accepting H_0 when it is false. (Referred as False Negative)
- We make Type-I error by rejecting H_0 when it is true. (Referred as False Positive)

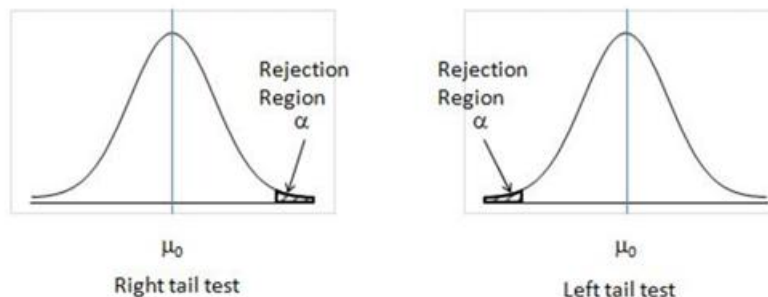


Controlling the significance level (α) helps minimize the risk of Type I errors, but it may increase the likelihood of Type II errors.

9. Number of Tails:

- **One-tailed test (directional)**

- Alternative hypothesis looks for change in **one specified direction only**, if observed mean significantly exceeds the hypothesized mean.
- **Right-tailed:** (used when checking larger values)
 - $H_0: \mu \leq \mu_0, H_1: \mu > \mu_0$
- **Left-tailed:** (used when checking smaller values)
 - $H_0: \mu \geq \mu_0, H_1: \mu < \mu_0$
- Rejection region is **only in one tail** (either upper or lower).



- **Two-tailed test (non-directional)**

- Alternative hypothesis looks for any change, either **increase or decrease**.
- Form:
 - $H_0: \mu = \mu_0$
 - $H_1: \mu \neq \mu_0$

- Rejection region is split between both tails of the sampling distribution (low and high extremes).

10. Conditions To Reject Null Hypothesis:

1. P-VALUE APPROACH:

A p-value is a probability that provides a measure of the evidence against the null hypothesis provided by the sample.

The **p-value** is the probability of obtaining a sample result at least as extreme as the observed result, assuming the null hypothesis H_0 is true.

How the p-value approach works

- Choose significance level α .
- Find the p-value (corresponding to test statistic, assuming null hypothesis is true).
- Decision: if **p-value $\leq \alpha$, reject H_0** ; if p-value $> \alpha$, do not reject H_0 .

Example:

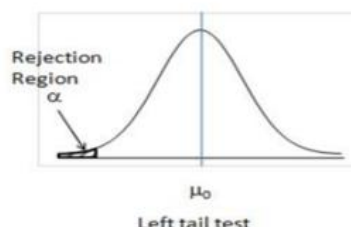
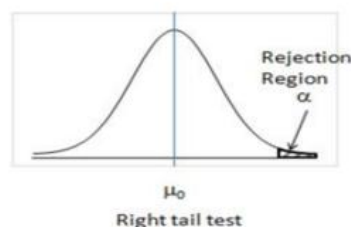
$t = 2.5 \rightarrow$ p-value = 0.012 : If $\alpha = 0.05$, then p-value (0.012) $< \alpha$ (0.05) $\rightarrow$ Reject H_0 .

2. CRITICAL VALUE APPROACH

Critical value is the value of test statistic that corresponds to an area of α i.e. Z_α .

How the Critical value approach works

- Set a critical value based on α .
 - Compute the test statistic and if it falls into the critical region (extreme enough) reject H_0
- For significance level $\alpha = 0.05$:
 - **For One-tailed test**
 - All α in one tail.
 - From standard statistical table, critical z-value ≈ 1.645 (right) or -1.645 (left) at 5% level.
 - Reject H_0 if: $z > 1.645$ (right-tailed test) Or $z < -1.645$ (left-tailed test)



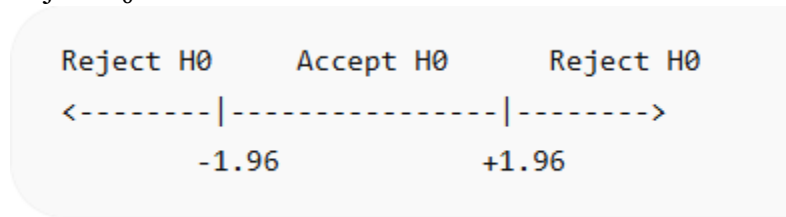
For the same α , a one-tailed test makes it “easier” to reject H_0 in the predicted direction (more power in that direction), but it completely ignores evidence in the opposite direction.

- **For Two-tailed test**

- Split α as $\alpha/2$ (0.025) in each tail.
- From the standard statistical table, the critical z-values cutting off 2.5% in each tail are:

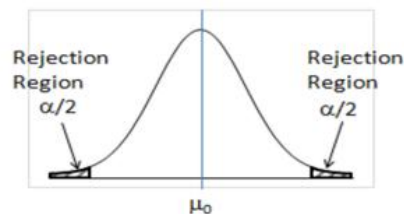
$$z \approx \pm 1.96$$

- Reject H_0 if: $z < -1.96$ or $z > 1.96$



$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$



- Since α is split, so at same significance level it is more difficult to reject null hypothesis in two-tailed test.

Example:

Suppose $\alpha = 0.05$ (one-tailed test), critical t-value = 1.645.

If calculated $t = 2.1$, then $2.1 > 1.645 \rightarrow$ falls in critical region $\rightarrow$ Reject H_0 .

The values of Z_α for most commonly used level of significance:

Value of Z_α	Level of Significance(α)					
	0.20	0.10	0.05	0.025	0.01	0.005
Z_α	0.842	1.282	1.645	1.960	2.326	2.576

Rejection rule using critical value

Lower tail test

Reject H_0

$$\text{if } Z \leq Z_\alpha$$

Upper tail test

Reject H_0

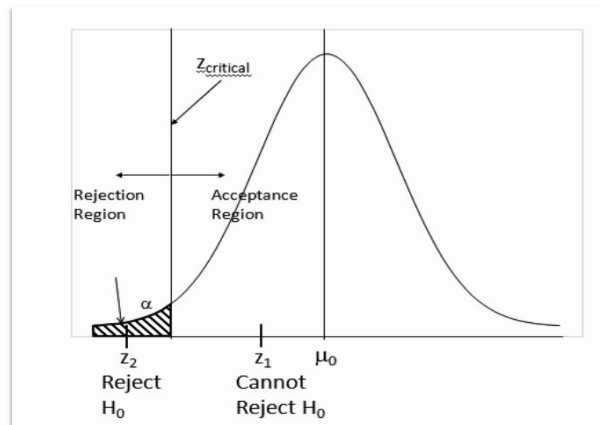
$$\text{if } Z \geq Z_\alpha$$

Two-tailed test

Reject H_0

$$\text{if } Z \leq -Z_{\alpha/2} \text{ or if } Z \geq Z_{\alpha/2}$$

Hypothesis Testing – Identifying Rejection Region



3.CONFIDENCE INTERVAL APPROACH

- Create a confidence interval (usually 95%).
- If the hypothesized value (under H_0) **falls outside** the confidence interval, **reject H_0** .

Example:

Suppose 95% CI for μ is (3.5, 4.5).

If H_0 claims $\mu = 5$, since 5 is not between 3.5 and 4.5 $\rightarrow$ Reject H_0 .

Why p-value approach and critical value approach give the same result

- Both start from the same setup: a null hypothesis, an alternative, a test statistic, and a fixed significance level α (e.g., 0.05).
- The critical-value approach defines a rejection region: if the test statistic falls beyond the critical value(s), you reject H_0 .
- The p-value approach computes the probability of seeing a test statistic at least as extreme as the observed one, assuming H_0 is true; if this p-value $\leq \alpha$, you reject H_0 . These two rules are mathematically equivalent:

$p\text{-value} \leq \alpha \Leftrightarrow \text{test statistic is in the rejection region defined by the critical value(s)}.$

- If the test statistic is beyond the critical value $\rightarrow$ that area is $\leq \alpha \rightarrow p\text{-value} \leq \alpha \rightarrow$ reject H_0 .
- If the test statistic is inside the acceptance region $\rightarrow$ that area is $> \alpha \rightarrow p\text{-value} > \alpha \rightarrow$ do not reject H_0 .

Imagine a **bell curve**:

- The **critical region** is at the tail(s) (extreme values). If your **test statistic** is in the critical region, **reject H_0** .
- The **p-value** tells how far in the tail your observed statistic is.

- The **confidence interval** tells the likely range of the true parameter — if H_0 's claim is outside, reject.

Which approach to consider:

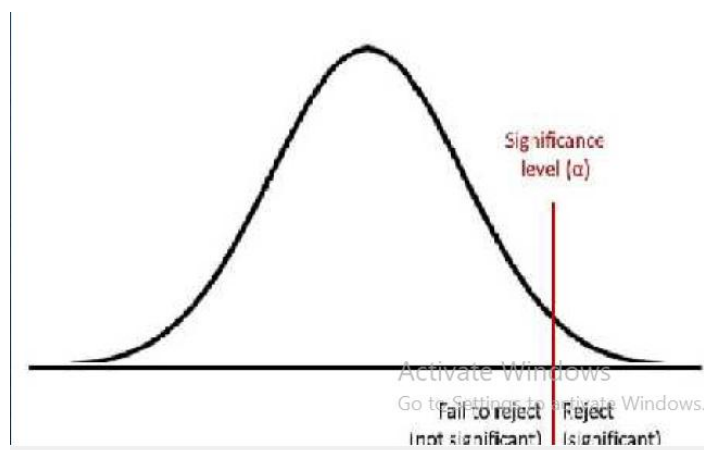
- **p-value** is **most common** in software (SPSS, R, Python, Excel).
- **Confidence intervals** are often used when you also want to **estimate** the effect size.
- **Critical regions** are **classical/manual** and good for understanding underlying logic.

Aspect	p-Value Approach	Confidence Interval Approach	Critical Region (Classical) Approach
Idea	Measures how likely your data would occur if H_0 is true.	Checks if the hypothesized value falls inside an interval estimate.	Compares test statistic to a critical value (threshold) to decide.
Decision Rule	Reject H_0 if $p\text{-value} \leq \alpha$.	Reject H_0 if hypothesized value is not in the confidence interval.	Reject H_0 if test statistic falls in critical region (e.g., beyond critical value).
Focus	Directly on the probability (p-value).	Range estimation for the parameter.	Standardized test statistic.
Example	If $p\text{-value} = 0.03$ and $\alpha = 0.05 \rightarrow$ reject H_0 .	If 95% CI is (3.2, 4.8) and H_0 claims $\mu = 5 \rightarrow$ reject H_0 (since 5 is not inside).	If $t = 2.2$ and critical value $t_{0.05} = 2.0 \rightarrow$ reject H_0 (since $2.2 > 2.0$).

The steps used for performing hypothesis testing is as follows:

1. **Formulate Hypothesis: Null (H_0) and alternative hypothesis (H_1).**
2. **Select Significance Level (α): Commonly used values are 0.05 or 0.01.**

The significance level (α), is a predetermined threshold **set by the researchers** before conducting hypothesis test. It represents the maximum acceptable probability of making Type I error, which is rejecting the null hypothesis when it is true. Commonly used significance levels are 0.05 (5%) and 0.01 (1%)



- Significance level (α): Threshold probability for rejecting H_0 . Typically 0.05 (5%)
- Confidence Level: $1 - \alpha$. If $\alpha = 0.05$, confidence level = 95%

- Critical Region: The range of values that leads to rejection of H_0

3. Select appropriate test : based on data type, sample size, known/unknown variance and distribution.

For the null hypothesis for 1 sample test, $H_0 : \mu = \mu_0$, where:

μ is population mean, μ_0 is hypothesized mean, we can choose any of 3 H_1 ,

Two-tailed test: checks for any difference ($\neq$) when $H_1 : \mu \neq \mu_0$ (split significance level)

Left-tailed test: checks if parameter is smaller than value ($<$) $H_1 : \mu < \mu_0$

Right-tailed test: checks if parameter is larger than value ($>$) when $H_1 : \mu > \mu_0$

TYPE OF TEST

WHEN TO USE

Z test

Large samples ($N > 30$), known population σ

t-test

small samples, unknown population σ

Chi-square

categorical data, goodness of fit

F-test

Comparing variance, ANOVA

4. Calculate test statistic Based on the sample data.

Test statistic is a number calculated from sample data during hypothesis testing. It quantifies how far is the sample result is from what the null hypothesis claims. (the discrepancy between the observed data and what would be expected under the null hypothesis.)

Like: t-statistics, z-score, chi-square statistic

Most test statistics follow this structure:

$$\text{Test Statistic} = \frac{\text{Observed Value} - \text{Expected Value}}{\text{Standard Error}}$$

For **Z-test**: $Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$

For **t-test**: $t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$

Where:

- $\bar{x}$ = sample mean

- μ_0 = hypothesized population mean
- σ = population standard deviation
- s = sample standard deviation
- n = sample size

5. Determine Critical Value(s)

Critical value is based on α and the type of test. Critical values for Z test for various significance level are:

Alpha	One Tail	Two Tail
0.10	1.28	± 1.645
0.05	1.645	± 1.96
0.01	2.33	± 2.576

- **Two-tailed:** Split α equally between both tails
- **One-tailed:** Place entire α in one tail

For Z test if $\alpha = 0.05$, $\alpha/2 = 0.025$, so $Z_{\text{critical}} = \pm 1.96$

Reject H_0 if $Z < -1.96$ or $Z > 1.96$

6. Compare test statistic to critical value or p-value to significance level and make a decision accordingly.

Method 1: Critical Value Approach

- **Reject H_0** if test statistic falls in critical region
- **Fail to reject H_0** if test statistic falls in acceptance region

Hypothesis testing	Test statistic	Rejection rule
Z-tests	Z-value	If Test statistic $\geq Z_{\text{crit}}$, then reject the null hypothesis. If Test statistic $\leq -Z_{\text{crit}}$, then reject the null hypothesis.
t-tests	t-value	If Test statistic $\geq t_{\text{crit}}$, then reject the null hypothesis H_0 . If Test statistic $\leq -t_{\text{crit}}$, then reject the null hypothesis H_0 .
ANOVA	F-value	If Test statistic $\geq F_{\text{crit}}$, then reject the null hypothesis H_0 .
Chi-square	chi-square	If Test statistic $\geq \chi_{\text{crit}}$, then reject the null hypothesis H_0 .

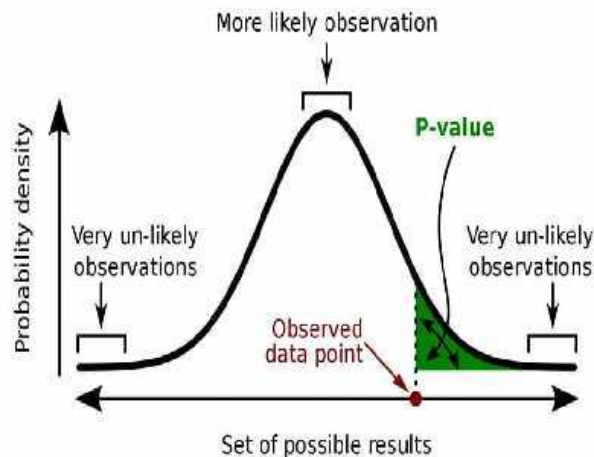
Method 2: P-value Approach

The p-value is a measure of the strength of evidence against null hypothesis based on the observed data.

- Calculate **p-value** (probability of observing test statistic or more extreme)
- **Reject H_0** if $p\text{-value} \leq \alpha$ (significance level),
- **Fail to reject H_0** if $p\text{-value} > \alpha$

Example:

- a. $P\text{-value} < 0.01$: Reject H_0 (statistically significant at 1% level)
- b. $P\text{-value} \geq 0.05$: Fail to reject H_0 (not statistically significant, there is insufficient evidence to reject null hypothesis)



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

7. Interpret the result in context of problem

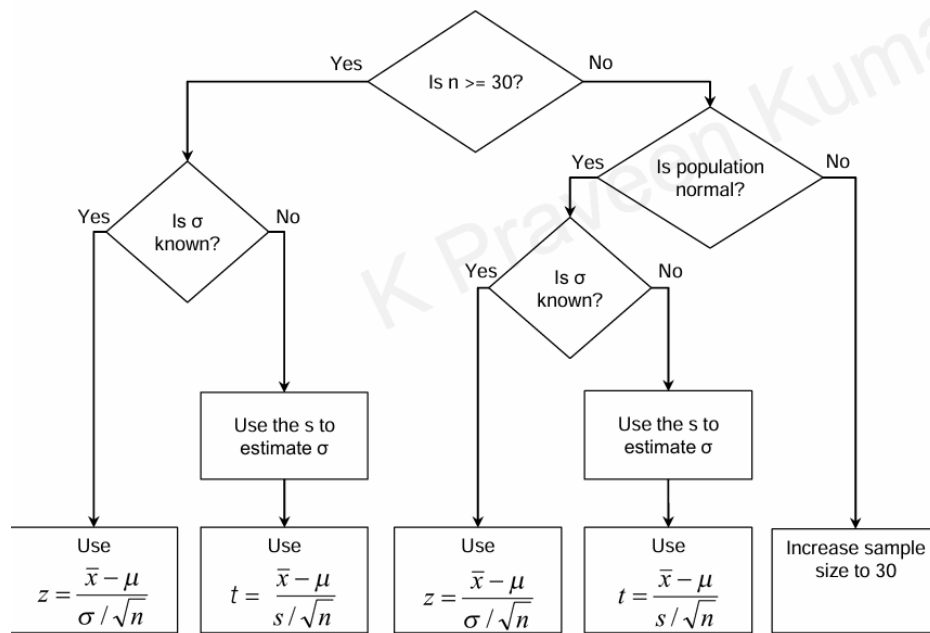
- **In statistical terms:** "We reject/fail to reject H_0 at the α significance level"
- **In practical terms:** Interpret what this means for the original problem

CCR Rule

- Calculate test statistic
- Compare with critical value
- **Reject** if beyond boundary

Hypothesis Testing Terminology:

- μ_0 : Hypothesized population mean.
- Significance level: $\alpha = 1 - \text{Confidence Level}$
- z : Observed value of $z = \left(\frac{\bar{x} - \mu_0}{\sigma} \right)$.
- z_{critical} : Value of z at the probability defined by significance level.
- Rejection region: Null Hypothesis is rejected if value of observed z lies in this region.



Problem: A company claims their light bulbs last 1000 hours on average. Test this claim.

1. **Hypotheses:**

- $H_0 : \mu = 1000$ hours
- $H_1 : \mu \neq 1000$ hours

2. **Significance Level:** $\alpha = 0.05$

3. **Test:** t-test (assuming small sample, unknown σ)

4. **Sample Data:** $n = 25$, $\bar{x} = 980$, $s = 50$

5. **Test Statistic:** $t = \frac{980-1000}{50/\sqrt{25}} = \frac{-20}{10} = -2.0$

6. **Critical Values:** ± 2.064 (for $df = 24$, $\alpha = 0.05$, two-tailed)

7. **Decision:** $|-2.0| < 2.064$, so **fail to reject H_0**

8. **Conclusion:** Insufficient evidence to conclude the mean differs from 1000 hours.

Introduction to t-Tests

A **t-test** is a statistical method used in Hypothesis testing **for comparison of the mean of the two normally distributed sample groups**. It determines whether there is a significant difference between means.

It is mainly used when:

- the sample size is small (< 30) and/or
- the population standard deviation σ is unknown and has to be estimated from sample standard deviation s .

The test uses the **t-distribution** instead of the normal z-distribution.

How to use t-test

Choose significance level and degree of freedom

Degree of freedom tells how many values in a calculation are free to vary while still satisfying a constraint. For many statistical tests, $df = n - 1$, as $n-1$ values can vary and one value gets fixed by condition.

- α : Typically 0.05 (5% risk of Type I error).
- **df** (affects critical values from t-distribution):
 - One-sample/paired: $df = n - 1$
 - Two-sample (equal variance): $df = n_1 + n_2 - 2$

Collect data and calculate t-test statistic

One Sample t-test: compares sample mean vs known hypothesised population mean.

Example: Is the average weight of a batch of medicine tablets equal to 500 mg?

Hypothesis:

$H_0: \mu = \mu_0$ (sample mean equals the hypothesized mean)

$H_1: \mu \neq \mu_0$ (two-tailed) OR $\mu > \mu_0$ OR $\mu < \mu_0$ (one-tailed)

Test Statistic Formula:

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

Where: $\bar{x}$ = sample mean, μ_0 = hypothesized population mean, s = sample standard deviation, n = sample size

Degrees of freedom: $df = n - 1$

Two Sample t-test: Compares means of two independent groups. Hypothesis is

Example: Comparing exam scores of students taught by Method A vs. Method B

Hypothesis:

$H_0: \mu_1 = \mu_2$ (no difference between group means)

$H_1: \mu_1 \neq \mu_2$

Test Statistic (Equal Variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (\text{pooled variance})$$

Degrees of freedom: $df = n_1 + n_2 - 2$

Paired t-test: Here we compare the means of two related or same group at two different time. (like change before and after treatment).

Hypothesis:

$H_0: \mu_d = 0$ (mean difference between pairs is zero)

$H_1: \mu_d \neq 0$

Test Statistic:

First, compute differences $d_i = x_{1i} - x_{2i}$, then: $t = \frac{\bar{d}}{s_d / \sqrt{n}}$ (n = number of pairs)

Where: $\bar{d}$ = mean of differences, s_d = SD of differences between each pair.

Degrees of freedom: $df = n - 1$

Determine critical value or p-value

- **Critical Value Approach:** Look up t-table for α , df, and tails (e.g., for two-tailed $\alpha=0.05$, $df=20$: ± 2.086).
 - Reject H_0 if $|t| > \text{critical value}$.
- **p-Value Approach** (preferred, via software like R/Python/Excel):
 - p-value = probability of observing $|t|$ or more extreme under H_0 .
 - Reject H_0 if p-value $\leq \alpha$.

Make conclusion

Reject H_0 : Evidence of significant difference ($p \leq \alpha$).

Fail to Reject H_0 : Insufficient evidence of difference ($p > \alpha$).

Interpret the result in context of problem

Worked Example:

A company claims their light bulbs last 1000 hours. A sample of 25 bulbs shows a mean life of 980 hours with SD = 60 hours. Test at $\alpha = 0.05$.

$$H_0: \mu = 1000; H_1: \mu \neq 1000$$

$$t = (980 - 1000) / (60 / \sqrt{25}) = -20 / 12 = -1.667$$

$$df = 24, \text{ critical } t = \pm 2.064$$

$$|-1.667| < 2.064, \text{ so we fail to reject } H_0$$

Conclusion: There is insufficient evidence that bulb life differs from 1000 hours.

Worked Example:

Group A (Drug): $n=20$, mean=75, SD=10. Group B (Placebo): $n=20$, mean=65, SD=12. Test at $\alpha = 0.05$.

$$H_0: \mu_1 = \mu_2; H_1: \mu_1 \neq \mu_2$$

$$sp^2 = (19 \times 100 + 19 \times 144) / 38 = 122, \text{ sp} = 11.05$$

$$t = (75 - 65) / 11.05 \sqrt{(1/20 + 1/20)} = 10 / 3.49 = 2.865$$

$$df = 38, \text{ critical } t = \pm 2.024$$

$$2.865 > 2.024, \text{ so reject } H_0$$

Conclusion: There is a significant difference between drug and placebo groups.

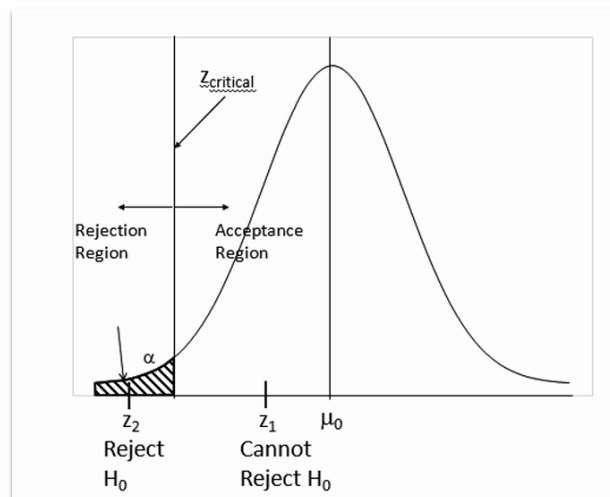
Worked Example:

Blood pressure measured before and after treatment in 10 patients. Differences (before-after): 5, 8, 3, 7, 10, 6, 2, 9, 4, 6. Mean difference = 6.0, SD = 2.54.

1. $t = 6.0 / (2.54 / \sqrt{10}) = 6.0 / 0.803 = 7.47$
2. $df = 9$, critical $t = \pm 2.262$ (at $\alpha=0.05$)
3. $7.47 > 2.262$, so reject H_0
4. Conclusion: Treatment significantly reduces blood pressure.

t-Test Type	Number of Groups	Groups Related?	Example Application
One-Sample	1 group vs. fixed value	N/A	Is product weight = 500g as labeled?
Independent Samples	2 groups	No (independent)	Male vs Female salary comparison
Paired Samples	1 group, 2 time points	Yes (same subjects)	Weight before and after diet program

Hypothesis Testing – Identifying Rejection Region



Z-test:

- A **z-test** is a statistical hypothesis test used to compare a **sample mean** with a **known population mean** (one-sample z-test), decide whether there is a “significant difference” between a sample and a population (or between two populations) when the sampling distribution of the statistic is approximately normal and the **population standard deviation is known**.

When is a z-test used?

- To compare a **sample mean** with a **known population mean** (one-sample z-test).
- To compare the **means of two independent samples**
- To test a **sample proportion** against a claimed population proportion (one-proportion z-test).

Z-test statistic (one-sample mean)

For testing whether a sample mean differs from a known population mean μ_0 :

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

where:

- $\bar{x}$ = sample mean,
- μ_0 = population mean under H_0 ,
- σ = **known** population standard deviation,
- n = sample size.

How to use z-test

1. **State hypotheses**
 - $H_0: \mu = \mu_0$ (no difference).
 - $H_1: \mu \neq \mu_0$, or $\mu > \mu_0$, or $\mu < \mu_0$ (two-tailed or one-tailed).
2. **Choose α** (e.g., 0.05) and find critical values from the **standard normal (z) table**:
 - For two-tailed $\alpha = 0.05$, critical values are approximately $z = \pm 1.96$.
3. **Compute the z-statistic** from the formula above.
4. **Decision rule**
 - **Critical-value approach**: reject H_0 if $|z| \geq z_{\text{critical}}$.
 - **p-value approach**: compute the two-sided (or one-sided) tail area under the standard normal; reject H_0 if $p\text{-value} \leq \alpha$.

Chi-Square Test

The **chi-square test** is a non-parametric statistical hypothesis test used to see whether categorical data differ significantly from what you would expect under a null hypothesis. Unlike t-tests which compare means, chi-square tests examine whether observed frequencies differ significantly from expected frequencies, or whether two categorical variables are associated. Chi-square is often written as χ^2 and is pronounced “kai-square”

It is mainly used for **frequency data** (counts in categories), not continuous measurements.

The generic chi-square statistic is:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where:

- χ^2 is the chi-square test statistic
- O is the observed frequency
- E is the expected frequency

For Chi-Square Tests

- Data are in the form of frequencies (counts), not percentages or proportions
- Observations are independent
- Expected frequency in each cell should be ≥ 5 (for validity)
- Categories are mutually exclusive

Chi-Square Goodness of Fit Test

Tests whether the observed frequency distribution of a categorical variable matches an expected distribution

Example:

- Testing if a die is fair (equal probability for each face)
- Does a sample distribution of blood types match national proportions?

Hypothesis:

H_0 : The observed frequencies match the expected distribution

H_1 : The observed frequencies do NOT match the expected distribution

Test Statistic:

$$\chi^2 = \sum [(O - E)^2 / E]$$

Where O = Observed frequency, E = Expected frequency

Degrees of freedom: $df = k - 1$ (k = number of categories)

Worked Example:

A six-sided die is rolled 120 times. Expected: 20 each. Observed: 18, 22, 17, 24, 19, 20.

$$\chi^2 = (18-20)^2/20 + (22-20)^2/20 + (17-20)^2/20 + (24-20)^2/20 + (19-20)^2/20 + (20-20)^2/20$$

$$\chi^2 = 0.2 + 0.2 + 0.45 + 0.8 + 0.05 + 0 = 1.70$$

$df = 5$, critical $\chi^2 = 11.07$ (at $\alpha=0.05$)

$1.70 < 11.07$, fail to reject H_0

Conclusion: The die appears to be fair.

Chi-Square Test of Independence

Tests whether two categorical variables are independent (no association) or dependent (associated) in a population.

When to Use:

- Examining if smoking status is related to lung cancer
- Testing if political affiliation is related to voting behavior
- Analyzing if treatment type is associated with recovery status

Hypotheses:

- H_0 : The two variables are independent (no association)
- H_1 : The two variables are dependent (association exists)

Constructing a Contingency Table:

	Outcome: Success	Outcome: Failure	Row Total
Treatment A	O_{11}	O_{12}	R_1
Treatment B	O_{21}	O_{22}	R_2
Column Total	C_1	C_2	N (Grand Total)

Expected Frequency Formula:

$$E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$$

Degrees of freedom: $df = (r-1)(c-1)$, where r = rows, c = columns

Worked Example:

Survey of 200 people: Gender vs. Product Preference.

	Prefers Product A	Prefers Product B	Total
Male	60 ($E=55$)	40 ($E=45$)	100
Female	50 ($E=55$)	50 ($E=45$)	100
Total	110	90	200

$$\chi^2 = (60-55)^2/55 + (40-45)^2/45 + (50-55)^2/55 + (50-45)^2/45$$

$$\chi^2 = 0.455 + 0.556 + 0.455 + 0.556 = 2.02$$

$$df = (2-1)(2-1) = 1, \text{ critical } \chi^2 = 3.841 \text{ (at } \alpha=0.05\text{)}$$

$$2.02 < 3.841, \text{ fail to reject } H_0$$

Conclusion: No significant association between gender and product preference.

Interpretation through Chi-Square test

- You compare **observed frequencies** (what you actually see in your data) with **expected frequencies** (what you would expect if the null hypothesis were true).
- If the observed counts are “too far” from the expected counts, i.e. larger the difference between the observations and expectations ($O - E$ in the equation), the test gives a **large chi-square value**, which suggests the null hypothesis may not be true.

To decide whether the difference is big enough to be statistically significant, you **compare the chi-square value to a critical value** with appropriate degrees of freedom and significance level (α) to decide whether to reject H_0 .

If Test statistic $\geq \chi_{crit}$, then reject the null hypothesis H_0 .

U-Test:

A U-test refers to **Mann-Whitney U test**, also called the **Wilcoxon rank-sum test**. It is a non parametric test used to compare two independent groups when:

- Data is not normally distributed and sample size is small
- Data is ordinal/ranked.

Instead of comparing numerical means assuming normality (like t-test), U test compares **the ranks of observations** across two groups, i.e. the data in one group tends to have consistently higher or lower values (ranks) than the other.

Example:

Suppose we want to compare reaction times (in seconds) of two independent groups.

Group A ($n_1=4$): 5, 7, 9, 10

Group B ($n_2=4$): 4, 6, 8, 12

Step1:

Null hypothesis (H_0): The distribution of reaction time in A and B are the same (no difference in ranks).

Alternative hypothesis (H_1): One group tends to have larger or smaller values than the other

Step2:

Combine and rank all observations.

Now assign ranks (smallest = 1):

- 4 → rank 1 (B)
- 5 → rank 2 (A)
- 6 → rank 3 (B)
- 7 → rank 4 (A)
- 8 → rank 5 (B)
- 9 → rank 6 (A)
- 10 → rank 7 (A)
- 12 → rank 8 (B)

- Group A values and ranks:
5 (2), 7 (4), 9 (6), 10 (7)
- Group B values and ranks:
4 (1), 6 (3), 8 (5), 12 (8)

$$R_1 \text{ for A: } 2 + 4 + 6 + 7 = 19$$

$$R_2 \text{ for B: } 1 + 3 + 5 + 8 = 17$$

- Find the sum of ranks of Group A and Group B:
- Compute U statistics using rank-sum formulas:

For $n_1=4$ (A), $n_2=4$ (B), $R_1=19$, $R_2=17$

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2$$

check $U_1 + U_2 = 7 + 9 = 16 = n_1 n_2$ (always true)

The Mann-Whitney U statistic used for testing is:

$$U = \min(U_1, U_2) = \min(7, 9) = 7$$

Interpretation: Based on this small sample, there isn't strong evidence that the two groups differ in reaction times.

If A and B are very mixed (no group systematically larger), U will be **around the middle** of its range.

If A is consistently larger than B, U for A will be **large** (and $U = \min(U_1, U_2)$ will be **small**), giving strong evidence against H_0 .

ANALYSIS OF VARIANCE (ANOVA)

ANOVA (Analysis of Variance) is used to test for significant differences among three or more group means simultaneously. Rather than conducting multiple t-tests (which inflates Type I error), ANOVA provides a single F-test to evaluate all group differences.

- If all population means are equal, the variability **between** group means should be small compared to the variability **within** groups.
- If at least one mean is different, the variability between groups will be **large** relative to within-group variability, giving a large F-statistic and a small p-value

Why Not Multiple t-Tests?

If we test 3 groups with 3 separate t-tests at $\alpha=0.05$, the probability of making at least one Type I error = $1 - (0.95)^3 = 14.3\%$. ANOVA controls this familywise error rate.

One-Way ANOVA

Tests the effect of one independent variable (factor) on a continuous dependent variable across three or more groups.

When to Use:

- Comparing test scores across 3 different teaching methods
- Comparing crop yields under 4 different fertilizer treatments
- Comparing reaction times across 5 age groups

Hypotheses:

$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$ (all group means are equal)

H_1 : At least one group mean is different from the others

ANOVA itself does **not** tell which groups differ, only that “some difference exists” overall.

Partitioning Variance (The Core Idea and Logic of F-statistics)

ANOVA splits total variability in the data into **between-group** and **within-group** parts.

1. **Total variability** (Total Sum of Squares, SST)
Measures how far each observation is from the **overall mean** $\bar{Y}$.
2. **Between-group variability** (Sum of Squares Between, SSB or SS_{between})
Measures how far each **group mean** $\bar{Y}_j$ is from the overall mean, weighted by group size.
3. **Within-group variability** (Sum of Squares Within, SSW or SS_{within})
Measures how spread out the data are **inside** each group around its own group mean.

These satisfy:

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}$$

Total Variation (SST) = Between-Group Variation (SSB) + Within-Group Variation (SSW)

Then you convert sums of squares to **mean squares** (variances) by dividing by their degrees of freedom:

- Mean Square Between:

$$MS_{\text{between}} = \frac{SS_{\text{between}}}{k - 1}$$

- Mean Square Within:

$$MS_{\text{within}} = \frac{SS_{\text{within}}}{N - k}$$

Where k = number of groups, N= total sample size

The F- Statistics is :

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

- If all group means are equal in the population, both mean squares estimate the same variance σ^2 , so F should be around 1.
- If some means differ, MS_{between} is inflated, so F becomes **greater than 1**. Large F leads to a small p-value.
 - Large F-value → strong evidence against H_0
 - Small F-value → group means are similar

Source	Sum of Squares (SS)	df	Mean Square (MS)	F-statistic
Between Groups	$SSB = \sum n_i(\bar{x}_i - \bar{x})^2$	k - 1	$MSB = SSB/(k-1)$	$F = MSB/MSW$
Within Groups	$SSW = \sum \sum (x_{ij} - \bar{x}_i)^2$	N - k	$MSW = SSW/(N-k)$	
Total	$SST = SSB + SSW$	N - 1		

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

ANOVA uses F-statistic:

Worked Example:

Three teaching methods tested on 5 students each. Scores: Method A: 70,75,80,65,70 (mean=72); Method B: 85,90,88,92,80 (mean=87); Method C: 60,65,70,55,65 (mean=63).

$$\text{Grand mean} = (72+87+63)/3 = 74$$

$$SSB = 5[(72-74)^2 + (87-74)^2 + (63-74)^2] = 5[4+169+121] = 1470$$

$$SSW \text{ (calculated from individual deviations)} \approx 380$$

$$F = (1470/2) / (380/12) = 735 / 31.67 = 23.2$$

$$\text{Critical } F(2,12) = 3.89 \text{ at } \alpha=0.05$$

$$23.2 > 3.89, \text{ reject } H_0$$

Conclusion: Teaching methods differ significantly in effectiveness.

Types of ANOVA

Once one-way ANOVA is clear, there are extensions.

- **One-way ANOVA**

One factor with k levels, one continuous response variable.

Used when there is:

- One independent variable (factor)
- One dependent variable

Example:

Comparing exam marks across 3 teaching methods.

Suppose marks are:

Method A	Method B	Method C
70	85	90
75	88	92
72	84	89

- **Two-way ANOVA (factorial ANOVA)**

Examines the effects of two independent variables (factors) on a dependent variable simultaneously, and tests for interaction between factors

- Two factors (e.g., Teaching method: A/B/C and Gender: M/F) affecting marks.

It tests:

- Main effect of factor A
- Main effect of factor B
- Interaction $A \times B$ (whether the effect of A depends on B).
-

ANOVA Summary Table (Two-Way):

Source	SS	df	MS	F
Factor A (main effect)	SSA	a - 1	MSA	MSA/MSE
Factor B (main effect)	SSB	b - 1	MSB	MSB/MSE
Interaction ($A \times B$)	SS(AB)	(a-1)(b-1)	MS(AB)	MS(AB)/MSE
Error (Within)	SSE	N - ab	MSE	

Source	SS	df	MS	F
Total	SST	N - 1		

4.5 Reporting ANOVA Results

Standard APA format for reporting ANOVA results:

One-way: $F(df_between, df_within) = value, p = value, \eta^2 = value$

Example: $F(2, 12) = 23.2, p < .001, \eta^2 = .79$ (large effect)

Effect Size: Eta-Squared (η^2)

$$\eta^2 = SSB / SST$$

$\eta^2 = 0.01 \rightarrow$ Small effect

$\eta^2 = 0.06 \rightarrow$ Medium effect

$\eta^2 = 0.14 \rightarrow$ Large effect

ADVANTAGES OF ANOVA

Compares many groups together

Reduces Type I error

Efficient statistical method

SCENARIO BASED QUESTIONS

Scenario 1:

A coffee shop claims that their average customer waits **less than 5 minutes** to be served. A customer doubts this and records a sample of wait times.

Steps:

- Hypotheses:**
 - $H_0: \mu = 5$ (Average wait time is 5 minutes)
 - $H_1: \mu > 5$ (Average wait time is **more** than 5 minutes)
- Significance Level:**
 - $\alpha = 0.05$
- Sample Data:**
 - Sample mean = 5.4 minutes, standard deviation = 1 minute, sample size = 30
- Test Statistic:**

Use **t-test** because sample size is small (<30 usually) or σ (population std. dev) is unknown.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{5.4 - 5}{1/\sqrt{30}} \approx 2.19$$

5 P-value:

From t-tables or software, p-value ≈ 0.018

6 Decision:

p-value (0.018) $< \alpha$ (0.05) $\rightarrow$ **Reject H_0**

7 Conclusion:

There is enough evidence to conclude that the average wait time is **more than 5 minutes**.

Scenario 2:

A company claims that **80%** of their customers are satisfied. A competitor suspects it's lower. A survey of 100 customers finds that only **72** are satisfied.

1. Hypotheses:

- $H_0: p = 0.80$
- $H_1: p < 0.80$

2. Significance Level:

- $\alpha = 0.05$

3. Sample Data:

- Sample proportion ($\hat{p}$) = $72/100 = 0.72$

4. Test Statistic: Use **z-test** for proportions.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.72 - 0.80}{\sqrt{0.8(0.2)/100}} \approx -2$$

5 P-value:

From z-tables, p-value ≈ 0.0228

6 Decision:

p-value (0.0228) $< \alpha$ (0.05) $\rightarrow$ **Reject H_0**

7 Conclusion:

There is evidence that customer satisfaction is **less than 80%**.

Scenario 3:

Left Tail Test

A consumer forum suspects that 300 gm pack of Narasu's Coffee, is underweight. The forum wants to check this by collecting a sample and using a 0.05 level of significance. Sample size is 36 and standard deviation is 18.019

SOLUTION:

$$H_0: \mu \geq 300(\mu_0)$$

$$H_a: \mu < 300(\mu_0)$$

Mu0	300
TAILS	LT
CL	95%
ALPHA	0.05
SSIZE	36
STAT	z
XBAR	297
SIGMA	18.01904
SE	3.003173
Z	-0.99894
ZCRIT	-1.64485
P-VALUE	0.158911

Using z-value:

This is a left tail test. We are looking for z value less than z critical (-1.645) to reject null hypothesis. Since z is greater than z critical we can not reject null hypothesis.

Using p-value:

Since in 1 Tail test, we obtained p-value (0.16) greater than alpha (0.05), we cannot reject null hypothesis.

Conclusion:

The claim that Narasu is underfilling can not be admitted.

Scenario 4:

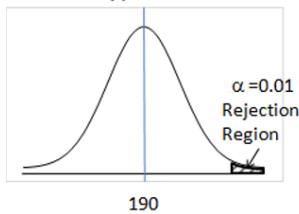
Right Tail Test

The Food Administration Authority received complaints that Maggi Noodle pack contains lead beyond permissible limits. The amount of lead should not exceed 190 gm. Conduct the Hypothesis test. Use a 0.05 level of significance. Sample size is 64 and Standard deviation is 16.003

Solution:

$$H_0: \mu \leq 190(\mu_0)$$

$$H_a: \mu > 190(\mu_0)$$



Mu0	190	Using z-value: This is a right tail test. We are looking for z value greater than z critical (1.645) to reject null hypothesis. Since z is greater than z critical we reject null hypothesis.
TAILS	RT	
CL	95%	Using p-value: Since in 1 Tail test, we obtained p-value(0.006) less than alpha(0.05), we reject null hypothesis.
ALPHA	0.05	
SSIZE	64	Conclusion: The claim that Maggi contains excess lead will be admitted.
STAT	z	
XBAR	195	
SIGMA	16.00397	
SE	2.000496	
Z	2.49938	
ZCRIT	1.644854	
P-VALUE	0.006221	

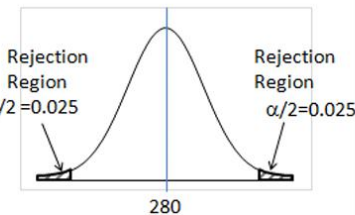
Scenario 5:

The weight of a drug has to be critically controlled. If less, the drug may not be effective, if more it may have some side effects. The manufacturer of the drug wants ascertain that the weight of the capsule is 280 mg at 95% confidence. Sample size is 36 and sigma is 12.00119

Mu0	280
TAILS	2T
CL	95%
ALPHA	0.05
ALPHA/2	0.025
SSIZE	36
STAT	z
XBAR	278.5
SIGMA	12.00119
SE	2.000198
Z	-0.74993
ZCRIT	-1.95996
P-VALUE	0.22665

$$H_0: \mu = 280(\mu_0)$$

$$H_a: \mu \neq 280(\mu_0)$$



Using z-value:

This is a two tail test. Observed z is negative. Therefore we apply Left Tail Test. Since z is greater than z critical we cannot reject null hypothesis.

Using p-value:

Since in 2 Tail test, we obtained p-value(0.22) greater than alpha(0.025), we cannot reject null hypothesis.

Conclusion:

The claim that Crocin contains paracetamol different from 280mg will not be admitted.

Scenario 6:

A purchase manger wants to check the diameter of aluminum die cast sourced from external supplier. The required diameter is 33 and is normally distributed. A sample was taken and is observed to have

$$n = 8$$

$$\bar{x} = 31.5$$

$$s = 1.3$$

Should the purchase manager accept the lot at 90% confidence?

Solution:

2 Tail Test

$$H_0: \mu = 33(\mu_0)$$

$$H_a: \mu \neq 33(\mu_0)$$

$$t = (31.5 - 33)/0.46 = -3.26$$

$$t_{crit} = TINV(0.1, 7) = -1.89$$

Decision: Reject H_0

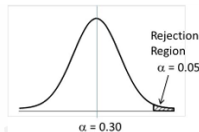
Scenario 7:

A marketing manager wants to introduce a new product. A blind comparison test is made with a sample size of 200 to assess customer acceptance of the product. The product will be launched only if favorable response is 30% plus. The product found favor with 64 respondents. Should the product be launched?

Solution:

$$H_0: p \leq .3 (\mu_0)$$

$$H_a: p > .3 (\mu_0)$$



C L	95%	σ (est)	0.458
Number of Tails	1	$z = (.32 - .3)/(.458/\sqrt{200})$	0.617
α	0.05	Z_{crit}	1.65
μ_0	0.3	p value	0.27
	0.32	Decision: Cannot reject H_0	